

Is the flow-to-flow problem well-posed?

Problem: Given edge flows, reconstruct new traffic equilibrium flows when the network is modified. OD-matrix is not given.

Well-posedness meaning:

- Does the problem have a solution? (Solution existence)
- Is the solution unique? (Solution uniqueness)

Answers:

- Existence: It is clear to see that a solution exists.
- Uniqueness: The solution is not unique, as there can be many different OD matrices generating the same pre-edit flows. When the network is edited these might yield different post-edit flows. **This statement might need a reference or some trivial example on a very small graph as motivation.**

Thus the deterministic flow-to-flow problem is not well-posed. We would need some other information, like a partial OD-matrix or some structural information about it.

However, the data-driven problem we are looking at seems well-posed. Fundamentally, we are aiming at estimating a conditional expectation.

Given OD-matrix $A \in \mathbb{R}^{|N| \times |N|}$ sampled from a distribution $p(A)$, the traffic-assignment function $f_{\mathcal{G}} : \mathbb{R}^{|N| \times |N|} \rightarrow \mathbb{R}^{|E|}$ assigns each edge $e \in E$ with a flow on the graph $\mathcal{G} = (N, E)$. We modify the original graph $\mathcal{G}$ into $\mathcal{G}'$. Then, the task is to obtain the estimate $\hat{F}$:

$$\hat{F}(f_{\mathcal{G}}(A), \mathcal{G}', \mathcal{G}) = \mathbb{E}_{A \sim p(A)} [f_{\mathcal{G}'}(A) \mid f_{\mathcal{G}}(A), \mathcal{G}'], \quad (1)$$

which is the unique minimizer of the MSE loss

$$\mathcal{L} = \mathbb{E}_{A \sim p(A), \mathcal{G}' \sim G'(\mathcal{G}), \mathcal{G} \sim G} \left[(f_{\mathcal{G}'}(A) - \hat{F}(f_{\mathcal{G}}(A), \mathcal{G}', \mathcal{G}))^2 \right].$$

Here, G is the distribution of the original graph, and $G'(\mathcal{G})$ is the distribution of the graph post-edit. Note that $f_{\mathcal{G}}(A)$ is a deterministic mapping given the graph and OD matrix, hence the only uncertainty is in A .

Obtaining the estimator in (1) is a well-posed problem. However, given that there is still uncertainty in the OD-matrix (we cannot recover A completely from $f_{\mathcal{G}}(A)$), the variance

$$\text{Var}(f_{\mathcal{G}'}(A) - \hat{F})$$

might still be quite high. There is simply no more information to use to reduce the variance. This could explain some of the relatively large errors we are seeing. As such, an interesting next step (it is not necessary to go down this route) would be to use additional information, for example if we know a couple of OD-pairs. It would also be interesting to go down the Bayesian path and try to obtain the uncertainty in the estimated flow.

Conclusion: The data-driven post-edit flow estimation problem is well-posed and interesting to study.